

Case Study on Major Incidents

Case Study No.: PNGRB/Tech/2026/11

Type of Installation: Gas Processing Plants/Units

No. of Fatality: 00

No. of Injury: 00

Incident Title : 181 minutes fire due to hydrocarbon leakage in Slug Catcher Area

Brief Description: An incident of hydrocarbon leakage followed by fire occurred in the slug catcher area of an oil and gas processing facility during excavation activities being carried out as part of a revamping project. The leakage originated from a small drain point connected to a live high-pressure gas pipeline.

At the time of the incident, an excavator was operating in close proximity to the pipeline. The excavator bucket struck the protruding drain point, resulting in sudden release of hydrocarbon. Multiple HC leak detection alarms were activated immediately, and evacuation of the affected area was initiated.

Within a few minutes, ignition occurred from a nearby operating unit, leading to a fire. Emergency response actions were promptly initiated, including isolation of the affected pipeline section, depressurisation, firefighting, and shutdown of adjoining units. The fire was contained within 30 mins and fully extinguished later. No injuries or fatalities were reported.



Observations/Lapses:

1. Excavation activity was undertaken in close proximity to a live high-pressure gas pipeline without adequate physical protection.
2. Protruding drain point connected to the pipeline was vulnerable to external mechanical impact and was not adequately safeguarded.
3. Heavy excavation equipment was allowed to operate near the pipeline without sufficient risk assessment or defined exclusion zone.
4. Permit to Work (PTW) issued for the activity was generic in nature and did not clearly identify the specific hazards associated with excavation near live pipelines.
5. Mandatory clearances from relevant technical disciplines prior to excavation were not effectively ensured.
6. Job Safety Analysis lacked clarity on individual responsibilities for implementation of control measures.
7. Adequate on-site supervision of contractor activities was not evident during execution of the work.

Root Causes:

1. **Inadequate risk assessment** and planning for excavation activities in a pipeline corridor.
2. **Non-compliance with permit-to-work** requirements and ineffective implementation of job safety controls.
3. **Absence of physical protection** for exposed pipeline drain point.
4. **Inadequate supervision** and control over contractor activities.
5. **Weak enforcement of safety management systems** during project execution.

Recommendations:

1. Excavation activities near live pipelines shall be permitted only after detailed risk assessment and implementation of adequate protective measures.
2. Physical guards, barriers, or encasements shall be provided to protect vulnerable pipeline appurtenances such as drain and vent points.
3. Use of heavy excavation machinery shall be restricted within defined buffer zones around live pipelines, and manual excavation shall be adopted where required.
4. Permit to Work system shall be strengthened to ensure permits are specific, hazard-focused, and strictly complied with.
5. Mandatory joint site survey involving operations, engineering, safety, and contractors shall be carried out prior to commencement of excavation activities.
6. Effective on-site supervision by responsible engineers shall be ensured during execution of high-risk jobs.
7. Contractors' personnel shall be adequately trained and sensitised regarding hazards of working in hydrocarbon installations.

Disclaimer:

➡ The Case Studies presented herein are based on real events and are intended for academic/learning purposes only. PNGRB thus disclaims all responsibility or liability for any costs, damages, losses, liabilities, expenses incurred by such third party arising out of or in connection with the report or any part thereto. This document does not aim to provide legal or any kind of expert advice; hence, the same shall not be relied upon before any authority or court of law.”

➡ The purpose of case studies sharing is strictly for knowledge dissemination and shared to promote learning & to enhance safety practices and to reduce the risk of similar incidents in the downstream Oil & Gas sector.

By reading this document, the reader is deemed to have agreed to the terms outlined herein.